Complete BCZ Farey Cycles: Rational Layers and Exact Least Periods

HCS-C395 source-theorem and reproducibility package

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Abstract

We reconstruct the complete periodic dynamics of the classical Boca–Cobeli–Zaharescu map on its half-open Farey triangle. A direct primitive-lattice-vector calculation proves the exact first-return map and excludes every irrational-slope periodic point. Each rational point has a unique positive scale and a reduced pair of integer coordinates. At each fixed scale these coordinates form one complete Farey denominator cycle, with least period equal to a totient sum. An elementary neighbor proof handles both endpoints of the Farey list without double counting, while the included upper and excluded lower scale boundaries are kept distinct. The map is an invertible area-preserving return with an exact coordinate-swap reversor. Independent exact controls compare floor iteration against separately sorted Farey fractions for all orders one through sixty-four. This is an explicitly classical, owner-heavy reconstruction, not a new-priority claim or a conversion of native denominator arithmetic into target prime data.

中文摘要

本文重建经典法里三角形返回映射的完整周期分类。原始格点向量的首次横向返回给出精确映射，
并排除全部无理斜率周期点。每个有理斜率点唯一确定实尺度和互素整数坐标；
固定尺度下，全部坐标组成一个完整法里分母循环，最小周期等于欧拉函数之和。
法里邻点证明明确处理首尾端点，不重复计数；尺度层的上端点包含、下端点排除。
交换坐标给出精确反演对称，归一化面积保持不变。 生成端与独立排序法里分数的检查端核对全部一至六十四阶有限循环。
这是明确承认经典归属的源系统重建，不把分母算术改名为目标素数数据。

Keywords: Farey triangle; horocycle return; primitive cycles; parabolic cocycle; roof integrability; source boundary
关键词：法里三角形；横流返回；本原循环；抛物矩阵；顶函数可积性；源系统边界

1 One section, native arithmetic, and classical ownership

The presence of exact arithmetic in a dynamical system does not by itself give a discrete prime-bearing orbit ledger. The Farey-triangle return is a particularly explicit test: its integer pairs are intrinsic primitive lattice coordinates, not labels added after the dynamics is computed. This paper keeps the source object, its clocks and its boundary conventions fixed throughout the argument.

The section and periodic classification are classical results of Athreya and Cheung [1]. Their construction identifies the Boca–Cobeli–Zaharescu map as a horocycle first return and relates its rational layers to Farey fractions. We reconstruct the needed arguments with explicit endpoint and matrix conventions and an independent finite verification package. This is an owner-heavy source reconstruction, not a claim of new literature priority. Cheung and Quas later established weak mixing [2]; that theorem is acknowledged as background and is neither reproved nor used below. Strong mixing and a complete spectral type are not claims of this paper.

Set

$$\Omega = \{(a,b) : 0 < a, b \le 1,\ a+b > 1\}, \qquad T(a,b) = (b, kb - a), \quad k = \left\lfloor \frac{1+a}{b} \right\rfloor. \tag{1}$$

The roof is $R(a,b) = 1/(ab)$. The floor takes its integer value at equality; the lower edge is excluded and the upper edges are included. We use column vectors, with branch matrix $A_k = \begin{pmatrix} 0 & 1 \\ -1 & k \end{pmatrix}$. One application of T is one discrete return, not one unit of physical horocycle time.

2 Exact first return, inverse, and irrational exclusion

Let

$$p_{a,b} = \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix}, \qquad h_s = \begin{pmatrix} 1 & 0 \\ -s & 1 \end{pmatrix}, \qquad \mathcal{L}_{a,b} = p_{a,b}\mathbb{Z}^2.$$

A lattice with a primitive positive horizontal generator of length $a \leq 1$ has a unique representation in this section: the second basis vector has height $1/a$, and its horizontal coordinate has a unique representative $b \in (1-a, 1]$ modulo $a\mathbb{Z}$. A lattice vector is $(x, n/a)$, where $x = am + bn$ and m, n are integers. At a positive horizontal-return time s, it obeys $0 < x \leq 1$ and $n/a = sx$, hence $n \geq 1$. If $s < 1/(ab)$, then $x > nb$, so $m \geq 1$ and $x \geq a + b > 1$, a contradiction. At $s = 1/(ab)$ the primitive vector $(b, 1/a)$ becomes $(b, 0)$. This proves the exact first return.

The new oriented basis is $(b, 0), (-a + kb, 1/b)$. The floor inequality gives $c = kb - a \leq 1 < c + b$, so $(b, c) \in \Omega$, including at equality. With $B_k = A_k^{\mathsf{T}}$, the basis identity is

$$p_{T(a,b)} = h_{R(a,b)} p_{a,b} B_k, \qquad B_k = \begin{pmatrix} 0 & -1 \\ 1 & k \end{pmatrix}. \tag{2}$$

We assert this first-return description for the stated section. We do not assert that this section exhausts every orbit of the whole lattice-space flow; the short closed horocycles omitted by the classical global section statement remain outside this claim.

If $T(a, b) = (b, c)$, then $(1 + c)/b = k + (1 - a)/b$ and $0 \leq (1 - a)/b < 1$. Thus

$$T^{-1}(a, b) = \left(\left\lfloor \frac{1 + b}{a} \right\rfloor a - b, a \right).$$

The swap $J(a, b) = (b, a)$ therefore satisfies $JTJ = T^{-1}$ and $R \circ J = R$, with the same endpoint convention. Every open branch has determinant one. Its countably many walls have area zero, and the inverse shows that branch images partition Ω up to this null set. Consequently $d\mu = 2\, da\, db$ is invariant probability.

Lemma 1. *Every periodic point of T has rational slope b/a.*

Proof. If $T^n(a, b) = (a, b)$ and $s_n = \sum_{j=0}^{n-1} R(T^j(a, b))$, iteration of (2) gives

$$B_{k_0} \cdots B_{k_{n-1}} = p_{a,b}^{-1} h_{-s_n} p_{a,b} = \begin{pmatrix} 1 - abs_n & -b^2 s_n \\ a^2 s_n & 1 + abs_n \end{pmatrix}. \tag{3}$$

The left side is integral. Hence $a^2 s_n$ is a positive integer and abs_n is an integer; their ratio is b/a. No continuity of the branch itinerary was used. $\square$

3 All rational layers and exact least periods

For $N \geq 1$, define

$$S_N = \{(q, r) : 1 \leq q, r \leq N,\ \gcd(q, r) = 1,\ q + r > N\}, \quad L_N = \sum_{j=1}^{N} \varphi(j), \quad \varphi(1) = 1.$$

Let $\mathcal{F}_N$ be the sorted reduced fractions in $[0, 1]$ with denominator at most N, including $0/1$ and $1/1$. There are $1 + L_N$ fractions, hence L_N gaps. Extend the list by integer translation when passing from the last gap to the first; the two endpoints are not two distinct marked points of the cyclic denominator list.

Lemma 2. *Consecutive denominator pairs in this cyclic list are exactly S_N, each once, and their successor is $(q, r) \mapsto (r, kr - q)$ with $k = \lfloor (N + q)/r \rfloor$.*

Proof. For determinant-one neighbors $p/q < s/r$, an intermediate fraction c/d satisfies

$$d = q(sd - rc) + r(qc - pd) \geq q + r.$$

Both parentheses are positive integers. This proves the absence of an intermediate denominator at most N whenever $q + r > N$. All neighbors have determinant one: induct from order one by inserting mediants exactly when the denominator sum is $N + 1$. Every new reduced c/d, $d = N + 1$, has these parents. Choose $q \in \{1, \dots, d - 1\}$ with $cq \equiv 1 \pmod{d}$, set $p = (cq - 1)/d$, $r = d - q$, and $s = c - p$. The parents lie in $[0, 1]$, have determinant one and denominator sum d, and the previous inequality proves they were neighbors at order N.

For a given $(q, r) \in S_N$, choose the unique $0 \leq p < q$ such that $pr \equiv -1 \pmod{q}$, setting $p = 0$ when $q = 1$. Then $s = (1 + pr)/q$ gives its unique neighboring fractions in $[0, 1]$. Finally $t = kr - q$ obeys $0 < t \leq N$ and $t + r > N$. The fraction $(ks - p)/t$ has determinant one with s/r; it is the next neighbor in the translated list. This proves the successor formula. $\square$

Theorem 1. *Every rational-slope point is uniquely $(\delta q, \delta r)$ with positive coprime q, r. For $N = \lfloor 1/\delta \rfloor$ its membership in Ω is equivalent to*

$$\frac{1}{N + 1} < \delta \leq \frac{1}{N}, \qquad (q, r) \in S_N.$$

At each such scale, δS_N is exactly one primitive cycle of least period L_N. These are all periodic cycles of the map.

Proof. Write $1/\delta = N + \theta, 0 \le \theta < 1$. Integer division gives

$$\left\lfloor \frac{1+\delta q}{\delta r} \right\rfloor = \left\lfloor \frac{N+q+\theta}{r} \right\rfloor = \left\lfloor \frac{N+q}{r} \right\rfloor,$$

since an integer remainder is at most $r-1$. Lemma 2 then visits every denominator pair exactly once. No earlier return is possible. The scale and the reduced pair are unique; Lemma 1 excludes any other periodic points. □

For clarity, $L_1 = 1$, $L_2 = 2$, $L_3 = 4$, and the third cycle is $(1,3), (3,2), (2,3), (3,1)$ before scaling. The lower scale endpoint belongs to the next order, not to this cycle. The map at the upper endpoint retains the exact floor values.

4 Independent evidence and scope

The canonical producer follows integer floor iteration from $(1, N)$. The independent checker imports no producer code: it constructs and sorts every reduced Farey fraction, compares every consecutive denominator pair, and verifies the floor inequalities, inverse recurrence and scalar lattice-return identity at an interior scale and at the included endpoint. All $N = 1, \dots, 64$ cycles contain 27,833 marked points in total. The two scales give 55,666 exact step controls. There are 27,833 starting-position return matrices, 320 positive repetition controls, 16 explicit floor-wall discontinuity controls and 128 fixed-iterate descriptions. These are complete finite populations, not an all-order proof by extrapolation.

The symbolic lane separately checks 14 universal matrix identities and 256 exact layer identities. Its 65 ninety-digit numerical controls are quadrature and asymptotic consistency checks, not certified enclosures. Two unrelated-directory producer replays are byte identical. Repaired-hash semantic attacks, strict literal-type and unknown-key checks, actual release-write rejection of malformed evaluation files, and refusal of optimized Python modes protect the declared evidence contract. Universal quantifiers are supplied by the proofs, not by these finite checks or by the release hash.

The source has intrinsic coprime arithmetic and complete primitive dynamics, but no natural correspondence from rational primes to primitive cycles of length $\log p$. Neither the totient sum nor the continuous physical roof supplies target amplitudes or signs. No target local arithmetic, Euler factors, root number, automorphy, divisor, functional equation, zero match or Hilbert–Polya operator is claimed. Route B remains disabled.

`NO_BAD_EULER_OR_ROOT_NUMBER`.

Round zero: complete rational layers and exact least periods.

References

[1] J. S. Athreya and Y. Cheung. A Poincaré section for the horocycle flow on the space of lattices. *International Mathematics Research Notices*, 2014(10):2643–2690, 2014. doi:10.1093/imrn/rnt003. Primary preprint arXiv:1206.6597.

[2] Y. Cheung and A. Quas. BCZ map is weakly mixing. arXiv:2403.14976, 2024.